Chapter 5

Public Key Encryption from Factoring

Our earlier encryption schemes were *symmetric*: the secret key needed to decrypt is also needed to encrypt. This can be a big problem, for example, when 1000 people want to be able to communicate pairwise. Instead, we will define and build a *public key* (also called asymmetric) scheme where the secret key is only needed to decrypt. Encryption requires a public key — public in the sense that it does not help the adversary mount an attack.

In this chapter, we will construct a public key encryption scheme and prove that, if factoring is hard, it enjoys a security property called *indistinguishability*.

Recall that there are typically three things that one must specify when defining a cryptographic scheme: syntax, correctness, and security. The **syntax** specifies what algorithms are required, along with their inputs and outputs. **Correctness** specifies what functionality the algorithms must satisfy when used by honest parties. **Security** specifies what guarantees are required in the face of an adversary.

5.1 Public Key Encryption: Syntax and Correctness

Compared to the definitions in Chapter 1, the definitions we present below reflect two big changes. The first, as mentioned above, is that we've changed the functionality we're after (public key vs symmetric encryption).

The second reflects a change in our approach to security, basing it on computational hardness assumptions using the formalisms of asymptotic analysis. We want to be able to tune the security of our scheme by increasing a *security parameter* $n \geq 1$. In general, the length of keys and ciphertexts (and possibly messages) will grow polynomially in the parameter n. This makes defining syntax and other properties a bit more complicated than in Chapter 1, where key length was generally fixed.

In particular, it makes it cumbersome to give explicit notation for the key-, message-, and ciphertext-spaces. We will usually leave the key space and ciphertext space implicit. For the message space, we will use $\mathcal{M} = \{\mathcal{M}_n\}$ to refer to an infinite collection of message spaces, one for each value of the security parameter n.

Definition 5.1 (Public Key Encryption (PKE): Syntax)**.** *A **public key encryption scheme (PKE)** for message space $\mathcal{M} = \{\mathcal{M}_n\}$ is a triple of PPT algorithms* $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *with inputs and outputs defined as follows. For security parameter $n \geq 1$:*

$\mathsf{Gen} : 1^n \mapsto (\mathsf{pk}, \mathsf{sk})$**.** *The* key generation *algorithm* Gen *takes as input the security parameter n written in unary. It outputs a* key pair $(\mathsf{pk}, \mathsf{sk})$ *where* pk *is called the* public key *and* sk *is called the* secret key*.*

$\mathsf{Enc} : (\mathsf{pk}, m) \mapsto c$**.** *The* encryption algorithm Enc *takes as input a public key* pk *and a message m. It outputs a ciphertext $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$.* Enc *may be randomized.*

$\mathsf{Dec} : (\mathsf{sk}, c) \mapsto m$**.** *The* decryption algorithm Dec *takes as input a secret key* sk *and a ciphertext c. It outputs a message $m = \mathsf{Dec}_{\mathsf{sk}}(c)$.* Dec *is deterministic.*

For notational simplicity, we assume the following:

- sk contains a copy of pk (or pk can easily be derived from sk). This is so Dec doesn't need to take pk as input.
- The length of pk is at least n bits. This is so algorithms given pk or sk as input (like Enc, Dec, and $\mathcal{A}$) can run in $\mathsf{poly}(n)$ time without needing the additional input 1^n.
- The message space $\mathcal{M}_n$ depends only on the security parameter n, and is independent of the keys. This is so we can say "for all $m \in \mathcal{M}_n$" before the keys are generated without running into technicalities.[1]
- It is efficient to check whether a given string x is a valid message: to verify that $x \in \mathcal{M}_n$ for a given x and n (and to recover n from pk). If not, $\mathsf{Enc}(\mathsf{pk}, x)$ returns $\perp$.

Correctness is defined much as before, with two minor changes. First, we require correctness to hold for all sufficiently large values of the security parameter n. Second, to quantify over all possible keys, we include the sampling of $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^n)$ in the probability experiment (instead of writing "for all $k \in \mathcal{K}$"). Because correctness must hold with probability 1, this implies that every possible key pair satisfies perfect correctness (i.e., there are no bad keys).

Definition 5.2 (PKE: Correctness)**.** *A PKE scheme* $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *is **correct** if for all sufficiently large n, and all $m \in \mathcal{M}_n$:*

$$\Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{Gen}(1^n), \\ \mathsf{Enc}}} \left[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m)) = m \right] = 1.$$

We require correctness only for all sufficiently large n (instead of for all n) because we don't care if the scheme is not defined / correct for small values of n. It is often the case that an encryption scheme we define in the natural way won't be defined for very small n. For example, the PKE in the following is not defined for $n = 1$ (because there are no 1-bit primes).

[1] This does not typically hold for the ciphertext space. There are natural encryption schemes where the message space depends on the specific public key pk, but that dependence can be removed without too much difficulty. See Example 5.2.

Example 5.1. Here is the OTP cast as a PKE for message space $\mathcal{M}_n = \{0,1\}^n$. It is correct. It is insecure because $\mathsf{pk} = \mathsf{sk}$ can be used to decrypt.

$\mathsf{Gen}(1^n)$: Sample $k \xleftarrow{\$} \{0,1\}^n$. Let $\mathsf{sk} = \mathsf{pk} = k$. Return $(\mathsf{pk}, \mathsf{sk})$.

$\mathsf{Enc}(\mathsf{pk}, m)$: Return $\mathsf{pk} \oplus m$

$\mathsf{Dec}(\mathsf{sk}, c)$: Return $\mathsf{sk} \oplus c$

Example 5.2. The following is a correct PKE scheme that is clearly insecure. It is defined for $n \geq 3$. The message space is $\mathcal{M}_n = \{1, \ldots, 2^{n-1}\}$.[2]

$\mathsf{Gen}(1^n)$:

- $p \leftarrow \mathsf{Primes}(1^n)$ `// uniform` n`-bit prime`
- $x \xleftarrow{\$} \mathbb{Z}_p^*$
- $\mathsf{pk} := (x, p)$
- $\mathsf{sk} := ([x^{-1} \bmod p], p)$
- Return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Enc}(\mathsf{pk}, m)$: Parse pk as (x, p). Return $[x \cdot m \bmod p]$

$\mathsf{Dec}(\mathsf{sk}, c)$: Parse sk as (s, p). Return $[s \cdot c \bmod p]$

To show that the scheme is correct, we check that:

$$\Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{Gen}(1^n), \\ \mathsf{Enc}}} \left[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m)) = m \right] = 1.$$

Plugging in the specific algorithms of the PKE scheme, we see that for $m \in \mathcal{M}_n$:

$$\begin{aligned} \mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m)) &= [[x^{-1} \bmod p] \cdot [x \cdot m \bmod p] \bmod p] \\ &= [x^{-1} x m \bmod p] \\ &= [m \bmod p] \\ &= m. \end{aligned}$$

The last equality is because $m \in \mathcal{M}_n$ implies $1 \leq m < p$.

It is insecure because a PPT adversary $\mathcal{A}$ given the public key $\mathsf{pk} = (x, p)$ can compute the secret key itself: $\mathsf{sk} = ([x^{p-2} \bmod p], p)$.

5.2 Rabin's PKE Scheme

For reference we recall our sampler for Blum integers (slightly modified to return p and q along with $N = pq$ for later notational convenience). $\mathsf{Primes}(1^n)$ samples a uniformly random n-bit prime.[3]

[2] If we didn't prefer the message space to be independent of the key, we could encrypt any message in $\mathbb{Z}_p^*$. Because p is n bits long, $\mathbb{Z}_p^*$ always contains $\mathcal{M}_n$.

[3] We require $n \geq 4$ for there to exist two distinct primes congruent to 3 mod 4.

$\mathsf{Blum}(1^n)$

$p \leftarrow \mathsf{Primes}(1^n)$, $q \leftarrow \mathsf{Primes}(1^n)$ independently
while $(q = p)$ *OR* $(p \not\equiv 3 \bmod 4)$ *OR* $(q \not\equiv 3 \bmod 4)$ **do**
  $p \leftarrow \mathsf{Primes}(1^n)$, $q \leftarrow \mathsf{Primes}(1^n)$ independently
$N := pq$
return (p, q, N)

Recall that modulo a Blum integer $N = pq$ ($p, q \equiv 3 \bmod 4$), every quadratic residue $y \in QR_N$ has exactly one square root r that is itself a quadratic residue $r \in QR_N$. This is called the *principal root*, and there is a PPT algorithm to compute it given the factorization p, q of N.

Definition 5.3. *For $N = pq$ a Blum integer and $y \in QR_N$, let $\mathsf{sqrtBlum}(y, p, q)$ be a deterministic polynomial-time algorithm returning the principal root of y modulo N.*

Definition 5.4 (Rabin's PKE)**.** *Rabin's PKE scheme* $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *for message space* $\mathcal{M}_n = \{0, 1\}$ *is defined as follows.*

$\mathsf{Gen}(1^n)$

$(p, q, N) \leftarrow \mathsf{Blum}(1^n)$
$\mathsf{sk} := (p, q)$
$\mathsf{pk} := N$
return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Enc}(\mathsf{pk}, m)$ **for** $\mathsf{pk} = N$

$r \xleftarrow{\$} QR_N$
$z := \mathsf{LSB}(r) \oplus m$
$y \leftarrow [r^2 \bmod N]$
return $c := (z, y)$

$\mathsf{Dec}(\mathsf{sk}, c)$ **for** $\mathsf{sk} = (p, q)$**,** $c = (z, y)$

$r := \mathsf{sqrtBlum}(y, p, q)$
return $m := \mathsf{LSB}(r) \oplus z$

Let's check correctness of Rabin's PKE.

Claim 5.1. *Rabin's PKE scheme is correct: for every sufficiently large n and every $m \in \{0, 1\}$,*

$$\Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{Gen}(1^n), \\ \mathsf{Enc}}} \left[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m)) = m \right] = 1.$$

Proof. Fix any key pair $(\mathsf{pk}, \mathsf{sk})$ generated by $\mathsf{Gen}(1^n)$, with $\mathsf{sk} = (p, q)$ and $\mathsf{pk} = N = pq$. Fix any message $m \in \{0,1\}$. Encryption samples $r \in QR_N$ and returns (z, y), where $z = \mathsf{LSB}(r) \oplus m$ and $y = [r^2 \bmod N]$. Since $r \in QR_N$ is a square root of y, it is the principal root of y and $\mathsf{sqrtBlum}(y, p, q) = r$. Therefore decryption returns

$$\mathsf{Dec}_{\mathsf{sk}}(z, y) = \mathsf{LSB}(r) \oplus z = \mathsf{LSB}(r) \oplus \big(\mathsf{LSB}(r) \oplus m\big) = m.$$

□

5.3 Ciphertext indistinguishability (IND-security) for 1-bit encryption

What security guarantee could we hope for when encrypting just 1 bit?

5.3.1 Perfect secrecy as a distinguishing experiment

In the symmetric encryption setting, *perfect indistinguishability* (equivalent to perfect secrecy) requires that for all ciphertexts $c \in \mathcal{C}$:

$$\Pr_{k \xleftarrow{\$} \mathcal{K}, \mathsf{Enc}}[\mathsf{Enc}_k(0) = c] = \Pr_{k \xleftarrow{\$} \mathcal{K}, \mathsf{Enc}}[\mathsf{Enc}_k(1) = c].$$

One implication of this is that no adversary $\mathcal{A}$ (even an unbounded $\mathcal{A}$ that is not PPT) can distinguish an encryption of 0 from an encryption of 1.

Consider the following probability experiment, still for symmetric encryption.

- $b \xleftarrow{\$} \{0,1\}$
- $k \xleftarrow{\$} \mathcal{K}$
- $c^* \leftarrow \mathsf{Enc}_k(b)$
- $\widehat{b} \leftarrow \mathcal{A}(c^*)$

Theorem 5.1
If Π is a perfectly secret symmetric encryption scheme, then for all $\mathcal{A}$ whose output always lies in $\{0,1\}$: $\Pr[\widehat{b} = b] = \frac{1}{2}$ in the above probability experiment.

Proof. By perfect indistinguishability, for any ciphertext c we can define

$$p_c := \Pr_{k, \mathsf{Enc}}[\mathsf{Enc}_k(0) = c] = \Pr_{k, \mathsf{Enc}}[\mathsf{Enc}_k(1) = c].$$

To evaluate $\Pr[\widehat{b} = b]$, we first separate the cases $b = 0$ and $b = 1$. That is:

$$\begin{aligned}\Pr[\widehat{b} = b] &= \Pr[b = 0] \cdot \Pr[\widehat{b} = b \mid b = 0] + \Pr[b = 1] \cdot \Pr[\widehat{b} = b \mid b = 1] \\ &= \frac{1}{2}\Big(\Pr[\widehat{b} = b \mid b = 0] + \Pr[\widehat{b} = b \mid b = 1]\Big).\end{aligned}$$

For $b = 0$:

$$\Pr[\widehat{b} = b \mid b = 0] = \sum_{c \in \mathcal{C}} \Pr[c^* = c \mid b = 0] \cdot \Pr[\mathcal{A}(c) = 0 \mid b = 0]$$

Observe that $\Pr[c^* = c \mid b = 0] = \Pr[\mathsf{Enc}_k(0) = c] = p_c$. The algorithm $\mathcal{A}$ only gets c as input. For every possible input c, there is some probability that $\mathcal{A}(c)$ returns 0. Let $a_c = \Pr[\mathcal{A}(c) = 0]$. Observe that, conditioned on its input c, $\mathcal{A}$'s behavior is independent of b. Therefore:

$$\Pr[\widehat{b} = b \mid b = 0] = \sum_{c \in \mathcal{C}} p_c a_c$$

For $b = 1$, by similar reasoning:

$$\begin{aligned}\Pr[\widehat{b} = b \mid b = 1] &= \sum_{c \in \mathcal{C}} \Pr[c^* = c \mid b = 1] \cdot \Pr[\mathcal{A}(c) = 1 \mid b = 1] \\ &= \sum_{c \in \mathcal{C}} p_c(1 - a_c).\end{aligned}$$

Putting it all together:

$$\Pr[\widehat{b} = b] = \frac{1}{2} \sum_{c \in \mathcal{C}} p_c\big(a_c + (1 - a_c)\big) = \frac{1}{2},$$

because $\sum_{c \in \mathcal{C}} p_c = 1$. □

5.3.2 Indistinguishability for PKE (1-bit messages)

An analogous statement to Theorem 5.1 doesn't hold for PKE. To make this precise, let's first define the probability that $\mathcal{A}$ is able to guess which message was encrypted in the experiment. In addition to the changes required by the syntax of PKE, there is one substantive change: **$\mathcal{A}$ gets pk as input.**

We define security in a modular way. First, we define a probability experiment $\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}$ capturing the following process. A public key pk is generated and used to create a **challenge ciphertext** encrypting a secret bit b: $c^* \leftarrow \mathsf{Enc}_{\mathsf{pk}}(b)$. $\mathcal{A}$ is given pk and c^* and outputs a guess $\hat{b}$ of the secret bit b.

We define $\mathcal{A}$'s indistinguishability advantage $\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}$ as its ability to beat random guessing, when $b \xleftarrow{\$} \{0,1\}$ is sampled uniformly at random. If $\mathcal{A}$ guessed randomly (or even always guessed $\hat{b} = 0$), it would be right with probability $1/2$. Security will require that this advantage is negligible.

> **Definition 5.5** (IND Security Experiment for PKE (simplified for 1-bit messages))**.** *Let Π be a PKE scheme for message space $\mathcal{M}_n = \{0,1\}$. For algorithm $\mathcal{A}$, $n \in \mathbb{N}$, and for $b \in \{0,1\}$, we define the following probability experiment:*

$\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,b)$ // Simplified for the special case of 1-bit messages

$(\mathsf{pk},\mathsf{sk}) \xleftarrow{\$} \mathsf{Gen}(1^n)$
$c^* \leftarrow \mathsf{Enc}_{\mathsf{pk}}(b)$
$\widehat{b} \leftarrow \mathcal{A}(\mathsf{pk}, c^*)$
return $\widehat{b}$

Definition 5.6 (IND Advantage for PKE)**.** *Let* $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *be a PKE scheme. For an algorithm* $\mathcal{A}$*, its* indistinguishability (IND) advantage *for* Π *is the following quantity:*

$$\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n) := \left|\Pr\left[\widehat{b} = b\right] - \frac{1}{2}\right|,$$

in the probability experiment

- $b \xleftarrow{\$} \{0,1\}$
- $\widehat{b} \leftarrow \mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,b)$

To see why an analogue of Theorem 5.1 can't hold, consider a 1-bit version of Example 5.1. It satisfies an analogue of the perfect indistinguishability condition (for $c \in \{0,1\}$):

$$\Pr_{\mathsf{pk} \xleftarrow{\$} \{0,1\}}[\mathsf{pk} \oplus 0 = c] = \frac{1}{2} = \Pr_{\mathsf{pk} \xleftarrow{\$} \{0,1\}}[\mathsf{pk} \oplus 1 = c].$$

But it is clearly insecure: $\mathcal{A}(\mathsf{pk}, c^*)$ can simply return $\widehat{b} = \mathsf{pk} \oplus c^* = \mathsf{Dec}_{\mathsf{sk}}(c^*)$ and achieve $\Pr[\widehat{b} = b] = 1$.

The step of the proof that breaks is the argument that $\mathcal{A}$ behaves the same whenever it is given a particular ciphertext c as input, regardless of whether c encrypts 0 or 1. This is no longer true because $\mathcal{A}$ also gets pk as input, not just c. Together, $\mathcal{A}$'s inputs (pk, c) uniquely determine the hidden bit b. This is true in general (not just for this example): it follows from correctness. (If there were two possible plaintexts for a given (pk, c), how could the decryptor know which one is right?) More generally, an unbounded adversary $\mathcal{A}$ given $(\mathsf{pk}, \mathsf{Enc}_{\mathsf{pk}}(m))$ can always recover m (see Exercise 5.4).

Despite the impossibility of PKE that is secure against an unbounded adversary, we can still hope for security against all PPT attackers. This motivates the following definition. It relaxes the previous definition in two ways: it allows *negligible* advantage (rather than requiring 0 advantage) and only for PPT algorithms (rather than all algorithms). On the other hand, it strengthens the previous definition by applying it to PKE, where the stronger definition is impossible to satisfy.

Definition 5.7 (IND security for PKE)**.** *PKE scheme* Π *is* IND secure *if for all PPT* $\mathcal{A}$*:*

$$\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n) \text{ is negligible.}$$

A first observation: **IND security requires encryption to be randomized**.

Theorem 5.2
If Π *is a correct PKE scheme with deterministic* Enc*, then* Π *is not IND secure.*

Proof. See Exercise 5.2 □

Definition 5.6 can be a little cumbersome to work with, because of the randomization over $b \xleftarrow{\$} \{0,1\}$ in the definition of $\mathbf{Adv}^{\mathsf{IND}}$. Much like in the proof of Theorem 5.1, one often has to apply Law of Total Probability to break into cases with $b = 0$ and $b = 1$. It is often easier to work with the following equivalent definition of security.

Definition 5.8 (IND security of PKE (equivalent alternative))**.** *PKE scheme* Π *for* $\mathcal{M}_n = \{0,1\}$ *is* IND secure *if for all PPT* $\mathcal{A}$*:*

$$\left|\Pr\left[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,0)=1\right] - \Pr\left[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,1)=1\right]\right| \text{ is negligible.}$$

That's a lot of notation that actually means something fairly simple. The expression above is equal to:

$$\left|\Pr\left[\mathcal{A}(\mathsf{pk}, \mathsf{Enc}_{\mathsf{pk}}(0)) = 1\right] - \Pr\left[\mathcal{A}(\mathsf{pk}, \mathsf{Enc}_{\mathsf{pk}}(1)) = 1\right]\right|$$

over the randomness of Gen, Enc and $\mathcal{A}$.

These two definitions are equivalent by the following claim:

Claim 5.2. *For all* Π*,* $\mathcal{A}$*,* n*:*

$$\left|\Pr\left[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,0)=1\right] - \Pr\left[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,1)=1\right]\right| = 2 \cdot \mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n).$$

Proof. See Exercise 5.3 □

5.4 IND-Security of Rabin's PKE

Theorem 5.3
If FACTORING*, then Rabin's PKE scheme is IND secure.*

To prove the theorem, we will show that a PPT adversary breaking Rabin's PKE implies a PPT algorithm for computing LSB of the (principal) square root of quadratic residues modulo N. By Theorem 4.3, this would violate FACTORING.

Proof. Let Π be Rabin's PKE. Suppose for contradiction (with the FACTORING assumption) there exists a PPT algorithm $\mathcal{A}$ such that

$$\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n) \text{ is not negligible.}$$

We will construct a PPT algorithm $\mathcal{B}$ for computing LSB such that

$$\left|\Pr\left[\widehat{b} = \mathsf{LSB}(x)\right] - \frac{1}{2}\right| = \epsilon(n) \tag{5.1}$$

for non-negligible ϵ in the probability experiment $(p, q, N) \leftarrow \mathsf{Blum}(1^n)$; $x \xleftarrow{\$} QR_N$; $y := [x^2 \bmod N]$; $\widehat{b} \leftarrow \mathcal{B}(N, y)$. By Theorem 4.3 (applied to $\mathcal{B}$ and its complement, see Exercise 4.1), this contradicts the FACTORING assumption. Therefore, no such PPT algorithm $\mathcal{A}$ can exist.

The key observation is this. Let $x \xleftarrow{\$} QR_N$ and $y = [x^2 \bmod N]$ be sampled as above. For any $z \in \{0,1\}$, the pair (z, y) is the output of $\mathsf{Enc}_{\mathsf{pk}}(m)$ using randomness x, for the message $m = z \oplus \mathsf{LSB}(x)$.

This suggests that if $\mathcal{B}(N, y)$ runs $\mathcal{A}$ on input $(N, (z, y))$ and $\mathcal{A}$ correctly recovers the message $m = z \oplus \mathsf{LSB}(x)$, then $\mathcal{B}$ can compute the LSB as $\hat{b} := m \oplus z = \mathsf{LSB}(x)$. To make this work, we must ensure that the distribution over $\mathcal{A}$'s input is such that we can invoke the fact that $\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n)$ is non-negligible. In particular, we want $\mathcal{A}$'s input to be distributed as an encryption of a uniformly random message. $\mathcal{B}$ can simulate this distribution by sampling z uniformly at random.

The above remarks suggest the following algorithm $\mathcal{B}$ for computing $\mathsf{LSB}(x)$ using $\mathcal{A}$ as a subroutine. The algorithm is clearly PPT whenever $\mathcal{A}$ is PPT.

$\mathcal{B}(N, y)$

$\mathsf{pk} := N$
$z \xleftarrow{\$} \{0,1\}$
$c^* := (z, y)$
$\widehat{b} \leftarrow \mathcal{A}(\mathsf{pk}, c^*)$ *// Interpret any output other than* 1 *as* 0.
return $\widehat{b} \oplus z$

By definition of $\mathcal{B}$: for all Blum integers N and $y \in QR_N$, with principal root x, and for z, c^* defined as in the code of $\mathcal{B}$:

$$\Pr\Big[\mathcal{B}(N, y) = \mathsf{LSB}(x)\Big] = \Pr\Big[\mathcal{A}(N, c^*) = \mathsf{LSB}(x) \oplus z\Big]$$

For N and y distributed as the probability experiment of equation (5.1), the right hand side is equal to:

$$= \Pr\Big[\mathcal{A}(\mathsf{pk}, c^*) = b\Big]$$

in the probability experiment: $b \xleftarrow{\$} \{0,1\}$; $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^n)$; $c^* \xleftarrow{\$} \mathsf{Enc}_{\mathsf{pk}}(b)$.

Therefore:

$$\left|\Pr\Big[\mathcal{B}(N, y) = \mathsf{LSB}(x)\Big] - \frac{1}{2}\right| = \left|\Pr\Big[\mathcal{A}(\mathsf{pk}, c^*) = b\Big] - \frac{1}{2}\right| = \mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n).$$

By assumption on $\mathcal{A}$, the above quantity is not negligible. Hence equation (5.1) holds, completing the proof.

□

5.5 Encrypting longer messages

We've done all this work just to encrypt a 1-bit message. Happily, a 1-bit PKE scheme that is IND secure implies a multi-bit PKE scheme that is also IND secure.

5.5.1 Indistinguishability for PKE for arbitrary message spaces

To generalize IND security beyond 1-bit messages, we want that $\mathcal{A}$ cannot distinguish $\mathsf{Enc}_{\mathsf{pk}}(m_0)$ from $\mathsf{Enc}_{\mathsf{pk}}(m_1)$ for every pair of messages m_0, m_1.

To formalize this, we need only update the probability experiment $\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}$ in Definition 5.5 to handle general message spaces. The rest of the definition—$\mathbf{Adv}^{\mathsf{IND}}_{\Pi,\mathcal{A}}$ and IND-security in Definitions 5.6 and 5.7—stays the same. This is the benefit of making our definitions modular.

The new version of the probability experiment works in two phases. In the first phase: the public key pk is generated and given to $\mathcal{A}$. Then $\mathcal{A}$ chooses a pair of messages (m_0, m_1) for which it thinks it can tell apart $\mathsf{Enc}_{\mathsf{pk}}(m_0)$ and $\mathsf{Enc}_{\mathsf{pk}}(m_1)$. In the second phase: one of the messages is encrypted to generate challenge ciphertext: $c^* \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m_b)$ for a bit $b \in \{0,1\}$. The challenge ciphertext is given to $\mathcal{A}$ which then outputs $\hat{b}$, its guess for the bit b.

There's an important wrinkle: as messages get longer, ciphertexts must necessarily get longer too.[4] See Exercise 5.5. This means that encryption can't completely hide the length of the messages. For this reason, we require that the messages m_0 and m_1 are the same length. If they are not, the experiment aborts by returning 0.

When the message space includes messages only of a fixed length, the check that $|m_0| = |m_1|$ always passes.

Definition 5.9 (IND Security Experiment for PKE)**.** *Let Π be a PKE scheme for message space $\mathcal{M} = \{\mathcal{M}_n\}$. For algorithm $\mathcal{A}$, $n \in \mathbb{N}$, and for $b \in \{0,1\}$, we define the following probability experiment:*

$\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{A}}(n,b)$ // For arbitrary message spaces

$(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{Gen}(1^n)$
$(m_0, m_1) \leftarrow \mathcal{A}(\mathsf{pk})$ *// Phase 1*
if $|m_0| \neq |m_1|$ *OR* $m_0 \notin \mathcal{M}_n$ *OR* $m_1 \notin \mathcal{M}_n$ **then**
  return 0
$c^* \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m_b)$
$\widehat{b} \leftarrow \mathcal{A}(\mathsf{pk}, c^*)$ *// Phase 2*
return $\widehat{b}$

Remark 5.1. We allow $\mathcal{A}$ to be **stateful**. For example, in Phase 2, $\mathcal{A}$ knows which messages (m_0, m_1) it returned in Phase 1 for free. We could instead make this explicit by adding a state variable that $\mathcal{A}$ outputs in Phase 1 and takes as input in Phase 2, but it makes things notationally more complex for little benefit.

5.5.2 From 1-bit PKE to multi-bit PKE

Let $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a PKE scheme for 1-bit messages ($\mathcal{M}_n = \{0,1\}$ for all n). We construct an encryption scheme $\Pi' = (\mathsf{Gen}', \mathsf{Enc}', \mathsf{Dec}')$ for ℓ-bit messages ($\mathcal{M}_n = \{0,1\}^\ell$ for all n).[5]

[4] If there is a maximum possible message length, then padding can be used to hide the length of messages.

[5] In fact, exactly the same construction can be used to encrypt arbitrary-length messages $\mathcal{M}_n = \{0,1\}^*$. We're avoiding doing so because it would require an additional layer of complexity to the proof.

$\mathsf{Gen}'(1^n)$

$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^n)$
return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Enc}'(\mathsf{pk}, m)$**, for** $m = m^1 m^2 \dots m^\ell$

for $i = 1, \dots, \ell$ **do**
  $c^i \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m^i)$ // *encrypt using fresh randomness*
$c := (c^1, c^2, \dots, c^\ell)$
return c

$\mathsf{Dec}'(\mathsf{sk}, c)$**, for** $c = (c^1, c^2, \dots, c^\ell)$

for $i = 1, \dots, \ell$ **do**
  $m^i \leftarrow \mathsf{Dec}_{\mathsf{sk}}(c^i)$
$m := m^1 m^2 \dots m^\ell$
return m

Theorem 5.4
Fix $\ell \geq 2$. Let Π be a PKE scheme for 1-bit messages and let Π' be the corresponding PKE scheme for ℓ-bit messages defined above. If Π is correct and IND-secure, then Π' is correct and IND-secure.

Proof Outline: The Hybrid Argument

The proof is going to use a new technique called a **hybrid argument**.

To prove IND security (Definition 5.8), we want to show that $\mathcal{A}$ behaves the same whether $c^* = \mathsf{Enc}'_{\mathsf{pk}}(m_0)$ or $c^* = \mathsf{Enc}'_{\mathsf{pk}}(m_1)$. That is:

$$\left| \Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n, 0) = 1] - \Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n, 1) = 1] \right| < \mathsf{negl}(n)$$

where $\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n, b)$ is the probability experiment where message m_b is encrypted.

What we will do is construct a sequence of intermediate experiments $\mathbf{Hyb}_j$ that are "hybrids" of these two possibilities, for $j = 0, \dots, \ell$. In the jth hybrid, we will generate the challenge ciphertext c^* by encrypting a hybrid message whose first j bits come from m_1 and whose remaining bits come from m_0.

The hybrids are designed to satisfy the following properties. Each adjacent pair of hybrids $\mathbf{Hyb}_j$ and $\mathbf{Hyb}_{j+1}$ differ in the encryption of a single bit. The first hybrid $\mathbf{Hyb}_0$ is identical to $\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n, 0)$ where $c^* \leftarrow \mathsf{Enc}'_{\mathsf{pk}}(m_0)$. The final hybrid $\mathbf{Hyb}_\ell$ is identical to $\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n, 1)$ where $c^* \leftarrow \mathsf{Enc}'_{\mathsf{pk}}(m_1)$.

Supposing for contradiction that Π' is not IND secure, there exists a PPT $\mathcal{A}$ such that

$$\left|\Pr[\mathbf{Hyb}_0 = 1] - \Pr[\mathbf{Hyb}_\ell = 1]\right|$$

is not negligible. Then we apply the Triangle Inequality.

$$\left|\Pr[\mathbf{Hyb}_0 = 1] - \Pr[\mathbf{Hyb}_\ell = 1]\right| \leq \sum_{j=0}^{\ell-1} \left|\Pr[\mathbf{Hyb}_j = 1] - \Pr[\mathbf{Hyb}_{j+1} = 1]\right|.$$

The crucial observation is this: if the left hand side is non-negligible, then at least one of the summands on the right hand side must also be non-negligible. This follows from an **averaging argument**: the maximum of a list of numbers $x_1, \ldots, x_k$ is at least as large as their average $(x_1 + \cdots + x_k)/k$.

The final step of the proof is to show that if

$$\left|\Pr[\mathbf{Hyb}_j = 1] - \Pr[\mathbf{Hyb}_{j+1} = 1]\right|$$

is non-negligible for some j, then Π is not IND secure. This is a contradiction. Hence Π' is IND secure.

Proof of Theorem 5.4

Correctness of Π' is easy to check. For all sufficiently large n, all $m \in \{0,1\}^\ell$ and all $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}'(1^n)$, the ith bit of $\mathsf{Dec}'_{\mathsf{sk}}(\mathsf{Enc}'_{\mathsf{pk}}(m))$ is equal to $\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m^i)) = m^i$, the ith bit of m. Because this holds for all i, $\mathsf{Dec}'_{\mathsf{sk}}(\mathsf{Enc}'_{\mathsf{pk}}(m)) = m$. Hence Π' is correct.

Suppose for contradiction that Π' is not IND secure. Then there exists a PPT $\mathcal{A}$ such that for some non-negligible function $\epsilon'(n)$:

$$\left|\Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n,0) = 1] - \Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n,1) = 1]\right| = \epsilon'(n)$$

Fix such $\mathcal{A}$. We may assume $\mathcal{A}$ always chooses $m_0, m_1 \in \{0,1\}^\ell$. If not, we can replace an invalid choice by $(0^\ell, 0^\ell)$ and force its final output to 0, preserving the modified experiment's output distribution.

For $j = 0, \ldots, \ell$, we define the hybrid probability experiment $\mathbf{Hyb}_j(n)$ as follows:

$\mathbf{Hyb}_j(n)$

$(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{Gen}(1^n)$
$(m_0, m_1) \leftarrow \mathcal{A}(\mathsf{pk})$
if $|m_0| \neq |m_1|$ *OR* $m_0 \notin \{0,1\}^\ell$ *OR* $m_1 \notin \{0,1\}^\ell$ **then**
  return 0
$m := m_1^1 m_1^2 \ldots m_1^j m_0^{j+1} m_0^{j+2} \ldots m_0^\ell$
$c^* \leftarrow \mathsf{Enc}'_{\mathsf{pk}}(m)$
$\widehat{b} \leftarrow \mathcal{A}(\mathsf{pk}, c^*)$
return $\widehat{b}$

By construction, $\mathbf{Hyb}_0(n) = \mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n,0)$ and $\mathbf{Hyb}_\ell(n) = \mathbf{Exp}^{\mathsf{IND}}_{\Pi',\mathcal{A}}(n,1)$. Therefore, by assumption on $\mathcal{A}$:

$$\left|\Pr[\mathbf{Hyb}_0 = 1] - \Pr[\mathbf{Hyb}_\ell = 1]\right| = \epsilon'(n)$$

By Triangle Inequality:

$$\epsilon'(n) \leq \sum_{j=0}^{\ell-1} \left|\Pr[\mathbf{Hyb}_j = 1] - \Pr[\mathbf{Hyb}_{j+1} = 1]\right|.$$

There must exist $0 \leq j < \ell$ such that

$$\left|\Pr[\mathbf{Hyb}_j = 1] - \Pr[\mathbf{Hyb}_{j+1} = 1]\right| \geq \frac{\epsilon'(n)}{\ell} =: \epsilon(n).$$

Because ℓ is polynomially-bounded (actually, constant), $\epsilon(n)$ is not negligible.

To complete the proof, we will show that the above implies Π (the 1-bit PKE) is not IND secure, yielding a contradiction. To do so, we construct a PPT adversary $\mathcal{B}_j$ attacking Π as follows.

$\mathcal{B}_j$

$\mathcal{B}_j(\mathsf{pk})$ *// Phase 1 of the IND security experiment*
  $(m_0, m_1) \leftarrow \mathcal{A}(\mathsf{pk})$
  return (m_0^{j+1}, m_1^{j+1})

$\mathcal{B}_j(\mathsf{pk}, c^*)$ *// Phase 2 of the IND security experiment, using* (m_0, m_1) *from Phase 1*
  for $i = 1, \ldots, j$ **do**
    $c_i \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m_1^i)$
  for $i = j+2, \ldots, \ell$ **do**
    $c_i \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m_0^i)$
  $c' = (c_1, \ldots, c_j, c^*, c_{j+2}, \ldots, c_\ell)$
  return $\mathcal{A}(\mathsf{pk}, c')$

$\mathcal{B}_j$ is PPT as long as $\mathcal{A}$ and Enc are PPT, which they are assumed to be.

By construction, $\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{B}_j}(n,0) = \mathbf{Hyb}_j$ and $\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{B}_j}(n,1) = \mathbf{Hyb}_{j+1}$. Therefore:

$$\left|\Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{B}_j}(n,0) = 1] - \Pr[\mathbf{Exp}^{\mathsf{IND}}_{\Pi,\mathcal{B}_j}(n,1) = 1]\right| = \left|\Pr[\mathbf{Hyb}_j = 1] - \Pr[\mathbf{Hyb}_{j+1} = 1]\right| \geq \epsilon(n)$$

for non-negligible $\epsilon(n)$.

This contradicts the IND-security of Π. Therefore our assumption that Π' is not IND secure is false, completing the proof. □

5.6 Exercise

Exercise 5.1 (Exercise 12.1 from Katz-Lindell)**.** Assume Π is a correct PKE scheme for 1-bit messages. Show that, given pk and a ciphertext c computed via $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$, it is possible for an unbounded adversary to determine m with probability 1.

Exercise 5.2. Prove Theorem 5.2, for general message spaces.

Exercise 5.3. Prove Claim 5.2, for general message spaces.

Exercise 5.4 (Based on Exercise 12.2 from Katz-Lindell)**.** Assume Π is a correct PKE scheme for 1-bit messages. Suppose that for all values of the security parameter n, ciphertexts are $L(n)$ bits long, for some $L(n) = O(\log n)$. Show that Π is not IND secure by describing a PPT adversary in pseudocode and analyzing its IND advantage.

Hint: The number of possible ciphertexts is $2^{L(n)}$ is polynomial in n.

Exercise 5.5 (Based on Exercise 3.2 from Katz-Lindell)**.** Let Π be a correct PKE scheme for message space $\mathcal{M}_n = \{0,1\}^*$.

(a) Show that there exists a polynomial $q(n)$ such that for all n, $|\mathsf{Enc}_{\mathsf{pk}}(0)| \leq q(n)$ and $|\mathsf{Enc}_{\mathsf{pk}}(1)| \leq q(n)$.

Hint: There is a very simple answer.

(b) Show that for $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^n)$ $m \xleftarrow{\$} \{0,1\}^{q(n)+2}$, and $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$

$$\Pr[|c| \leq q(n)] < 0.99.$$

Exercise 5.6. When defining IND security, we stated that Definition 5.5 is a simplified version of Definition 5.9 which can be used when the message space is $\mathcal{M}_n = \{0,1\}$. But syntactically, the security experiments are quite different. For example, in the simplified definition the adversary has just 1 phase; in the general definition the adversary has 2. This problem shows that the induced definitions of IND security for 1-bit messages coincide.

Let Π be a PKE scheme for 1-bit messages $\mathcal{M}_n = \{0,1\}$. For PPT algorithm $\mathcal{A}$, define the following quantities:

- Let $\mathbf{Adv}^{\mathsf{s}}_{\pi,\mathcal{A}}(n)$ be the IND advantage (Definition 5.6) defined using the **S**implified experiment Definition 5.5.
- Let $\mathbf{Adv}^{\mathsf{g}}_{\pi,\mathcal{A}}(n)$ be the IND advantage (Definition 5.6) defined using the **G**eneral experiment Definition 5.9.

(a) Given an arbitrary PPT algorithm $\mathcal{S}$, describe in pseudocode a PPT algorithm $\mathcal{G}$ such that

$$\mathbf{Adv}^{\mathsf{g}}_{\Pi,\mathcal{G}}(n) = \mathbf{Adv}^{\mathsf{s}}_{\Pi,\mathcal{S}}(n).$$

Justify your answer.

(b) ★ Suppose that for PPT $\mathcal{G}$:

$$\left|\mathbf{Adv}^{\mathsf{g}}_{\Pi,\mathcal{G}}(n) - \frac{1}{2}\right| = \epsilon(n) \text{ is not negligible.}$$

Prove that there exists a PPT $\mathcal{S}$ such that

$$\left|\mathbf{Adv}^{\mathsf{s}}_{\Pi,\mathcal{S}}(n) - \frac{1}{2}\right| \text{ is not negligible.}$$